De-multiplexing of Address/Data Bus (Self-Learning)

Q. Why does the 8086 use a multiplexed Address/Data bus? Explain the process of de-multiplexing this bus using Address Latch Enable (ALE) and Latches.

> **Definition to Remember**

> To s
during
**ALE*

---

1. Need for Multiplexing and De-multiplexing

- **Multiplexing:** The 8086 needs 20 pins for the address and 16 pins for data. Having 36 separate pins on a 40-pin chip leaves no room for power, clock, and control signals. Hence, they are multiplexed.

- **
addres

2. The Role of ALE (Address Latch Enable)

- The 8086 generates a positive pulse on the **ALE** pin strictly during the **T1 state** of every machine cycle.

- Thi

3. Using Latches (74LS373) for De-multiplexing

To de-multiplex, external **D-type transparent latches** (e.g., 74LS373) are used.

- **H

> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

> `Mul
Addres